

Comprehensive Pharmacogenetic & Pharmacodynamic Profile for Brian

Patient Factors:

- Personal history: Former alcoholic, history of anxiety & panic attacks
- Family history:
 - Grandfather & mother – Alcoholism (suggesting genetic predisposition to dopamine & GABA dysregulation)
 - Father – Anxiety & OCD (suggesting heightened serotonin/norepinephrine sensitivity)
- Profession: Electrical Engineer at Boeing (suggesting high cognitive demand, analytical mindset, potential for perfectionism & stress-related anxiety)
- Allergies: Tree nuts, particularly pistachios (potential concern for medication excipients)

1. Genetic Predisposition & Neurotransmitter Dysfunction

- ♦ Alcoholism in Family & Self → Likely Dopamine & GABA Dysregulation
 - Chronic alcohol use depletes dopamine, leading to low motivation, emotional numbness, and cravings for stimulation (like Wellbutrin).
 - GABA receptor downregulation → Increased baseline anxiety and reactivity after long-term alcohol use.
 - Family history of alcoholism suggests a genetic predisposition to dopamine dysregulation (possibly low baseline dopamine tone).
 - If he has the DRD2 A1 allele, he may struggle with dopamine reward sensitivity, making him more prone to impulsivity, OCD tendencies, or substance-seeking behaviors.
- ♦ Father's Anxiety & OCD → Heightened Serotonin/Norepinephrine Sensitivity
 - Genetic OCD link suggests serotonin transporter (SLC6A4) variations, making SSRI/SNRI response variable.
 - COMT "Worrier" (Met/Met) genotype likely → Slow dopamine breakdown, leading to overthinking, perfectionism, and higher norepinephrine sensitivity (panic-prone).

 Prediction:

🔴 Lower dopamine receptor density (due to genetic & alcohol history) → More prone to dopamine-seeking behaviors but also stimulant sensitivity.

🔴 Heightened serotonin/norepinephrine reactivity → More sensitive to anxiety-related side effects from Wellbutrin & Citalopram.

🔴 GABA dysfunction → Increased susceptibility to panic & difficulty calming autonomic nervous system.

2. Pharmacogenetic & Pharmacodynamic Concerns

Citalopram (Celexa) & Propranolol – Poor Fit for Brian

🔴 CYP2C19 Poor Metabolizer → Citalopram Buildup → Increased Anxiety & QT Prolongation Risk

- Citalopram relies on CYP2C19 for metabolism, and if he is a poor metabolizer, the drug accumulates.
- This increases serotonin overstimulation, leading to restlessness, panic, and autonomic hyperreactivity.
- Citalopram + Propranolol interaction → Inhibits CYP2D6, increasing propranolol levels, possibly lowering blood pressure too much and exacerbating dizziness/fatigue.

Wellbutrin (Bupropion) – Overstimulating for Him

🔴 CYP2D6 Inhibitor → Increased Propranolol Effects → Exaggerated Sympathetic Activation

- Wellbutrin blocks CYP2D6, slowing propranolol metabolism, making propranolol effects stronger, possibly causing a paradoxical adrenaline rebound effect and increasing panic episodes.
- Wellbutrin also increases norepinephrine and dopamine, which could have led to agitation and worsening throat tightness.

Throat Tightness – Allergy or Neurochemical Sensitivity?

🔴 Tree Nut Allergy (Anaphylaxis Risk) → Wellbutrin Fillers Could Be a Problem

- Some formulations of Wellbutrin contain potential excipients derived from tree nuts (e.g., pistachio derivatives in certain extended-release formulations).
- However, throat tightness can also be a neurological autonomic response from serotonin & norepinephrine overstimulation.

✅ Predicted Cause of Throat Tightness:

- Combination of allergic potential + pharmacodynamic overstimulation from Wellbutrin & Citalopram.

3. The Gut-Brain Axis & How Alcohol + Omeprazole Are Making Anxiety Worse

♦ Alcohol-Induced Gut Damage → Neurotransmitter Imbalance

- Alcohol reduces gut microbiota diversity, leading to lower serotonin and GABA levels (exacerbating anxiety & panic symptoms).
- Chronic gut inflammation disrupts the gut-brain axis, contributing to mood swings, irritability, and cognitive fog.

♦ Omeprazole (Prilosec) & Nutrient Deficiencies → Worsening Neurotransmitter Production

B12 Deficiency → Low Dopamine & Serotonin → Worsening Fatigue & Depression

- B12 is crucial for dopamine synthesis, and low B12 levels contribute to cognitive slowing, low motivation, and anxiety.

Folate Deficiency → Impaired Serotonin & Norepinephrine Regulation

- Low folate leads to worsened mood instability and heightened stress sensitivity.

Vitamin D3 Deficiency → Increased Depression & Anxiety

- Vitamin D helps regulate dopamine & serotonin, and low levels increase panic vulnerability.

Plan:

- Switch from Omeprazole to Pantoprazole to reduce nutrient malabsorption.
- Correct B12, Folate, and Vitamin D3 deficiencies.
- Address gut health with probiotic & anti-inflammatory support.

4. Why Duloxetine (Cymbalta) Is the Best Fit for Him

Serotonin + Norepinephrine Modulation Without Excess Dopamine Stimulation

- Duloxetine provides a balanced increase in serotonin & norepinephrine without the dopamine spikes that triggered panic on Wellbutrin.
- Less cardiac risk compared to Citalopram.
- Less inhibition of propranolol metabolism, avoiding unpredictable adrenaline surges.

✓ Helps Regulate Autonomic Nervous System Instability (Throat Tightness, Panic)

- Duloxetine stabilizes norepinephrine transmission, reducing panic symptoms, throat tightness, and autonomic dysregulation.

✓ Supports Gut-Brain Health & Alcohol Recovery

- Addresses serotonin depletion from past alcohol use.
- Does not interfere with gut absorption like Omeprazole.

✓ Reduces Risk of Drug Interactions & Overstimulation

- No significant QT prolongation risk (safer for cardiac health).
- Less inhibition of CYP2D6, making propranolol more predictable in its effects.

5. Personalized Cross-Taper Plan

Week 1:

- Lower Citalopram to 30 mg
- Start Duloxetine 30 mg
- Keep Wellbutrin at 150 mg

Week 2:

- Lower Citalopram to 20 mg
- Increase Duloxetine to 60 mg

Week 3:

- Lower Citalopram to 10 mg
- Stop Wellbutrin completely

Week 4:

- Stop Citalopram completely
- Continue Duloxetine 60 mg

♦ Additional Adjustments:

- Switch Omeprazole to Pantoprazole.

- Correct B12, folate, and vitamin D deficiencies.
- Monitor autonomic symptoms (throat tightness, heart rate, panic) during transition.

Final Thoughts: Why This Plan Works for Brian

- ♦ Pharmacogenetics: Avoids drugs he metabolizes poorly (Citalopram, Wellbutrin).
- ♦ Anxiety Control: Duloxetine provides steady serotonin & norepinephrine balance without overstimulation.
- ♦ Autonomic Stability: Less adrenaline surges, fewer cardiac risks, no unpredictable propranolol interactions.
- ♦ Nutrient Support: Prevents gut-brain deficiencies from PPI use & past alcohol history.

This plan is customized to work with your unique neurochemistry, metabolic profile, and history for long-term emotional resilience and stability.

Best,

Anna Tran, PMHNP-BC

LifeStance Health